

Key guidance derived from Chapter II:

- ✓ Ensure that the AI system's entire life cycle meets the seven key requirements for Trustworthy AI: (1) human agency and oversight, (2) technical robustness and safety, (3) privacy and data governance, (4) transparency, (5) diversity, non-discrimination and fairness, (6) environmental and societal well-being and (7) accountability.
- ✓ Consider technical and non-technical methods to ensure the implementation of those requirements.
- ✓ Foster research and innovation to help assessing AI systems and to further the achievement of the requirements; disseminate results and open questions to the wider public, and systematically train a new generation of experts in AI ethics.
- ✓ Communicate, in a clear and proactive manner, information to stakeholders about the AI system's capabilities and limitations, enabling realistic expectation setting, and about the manner in which the requirements are implemented. Be transparent about the fact that they are dealing with an AI system.
- ✓ Facilitate the traceability and auditability of AI systems, particularly in critical contexts and situations.
- ✓ Involve stakeholders throughout the AI system's life cycle. Foster training and education so that all stakeholders are aware of and trained in Trustworthy AI.
- ✓ Be mindful that there might be fundamental tensions between different principles and requirements. Continuously identify, evaluate, document and communicate these trade-offs and their solutions.

III. Chapter III: Assessing Trustworthy AI

Based on the key requirements of Chapter II, this Chapter sets out a non-exhaustive **Trustworthy AI assessment list** (pilot version) to **operationalise Trustworthy AI**. It particularly applies to AI systems that directly interact with users, and is primarily addressed to developers and deployers of AI systems (whether self-developed or acquired from third parties). This assessment list does not address the operationalisation of the first component of Trustworthy AI (lawful AI). Compliance with this assessment list is not evidence of legal compliance, nor is it intended as guidance to ensure compliance with applicable law. Given the application-specificity of AI systems, the assessment list will need to be tailored to the specific use case and context in which the system operates. In addition, this chapter offers a general recommendation on how to implement the assessment list for Trustworthy AI through a governance structure embracing both operational and management level.

The assessment list and governance structure will be developed in close collaboration with stakeholders across the public and private sector. The process will be driven as a piloting process, allowing for extensive feedback from two parallel processes:

- a) a qualitative process, ensuring representability, where a small selection of companies, organisations and institutions (from different sectors and of different sizes) will sign up to pilot the assessment list and the governance structure in practice and to provide in-depth feedback;
- b) a quantitative process where all interested stakeholders can sign up to pilot the assessment list and provide feedback through an open consultation.

After the piloting phase, we will integrate the results from the feedback process into the assessment list and prepare a revised version in early 2020. The aim is to achieve a framework that can be horizontally used across all applications and hence offer a foundation for ensuring Trustworthy AI in all domains. Once such foundation has been established, a sectorial or application-specific framework could be developed.

■ *Governance*

Stakeholders may wish to consider how the Trustworthy AI assessment list can be implemented in their organisation. This can be done by incorporating the assessment process into existing governance mechanisms, or by implementing new processes. This choice will depend on the internal structure of the organisation as well as its size and available resources.

Research demonstrates that management attention at the highest level is essential to achieve change.⁵⁵ It also demonstrates that involving all stakeholders in a company, organisation or institution fosters the acceptance and the relevance of the introduction of any new process (whether or not technological).⁵⁶ Therefore, we recommend implementing a process that embraces both the involvement of operational level as well as top management level.

Level	Relevant roles (depending on the organisation)
Management and Board	Top management discusses and evaluates the AI systems' development, deployment or procurement and serves as an escalation board for evaluating all AI innovations and uses, when critical concerns are detected. It involves those impacted by the possible introduction of AI systems (e.g. workers) and their representatives throughout the process via information, consultation and participation procedures.
Compliance/Legal department/Corporate responsibility department	The responsibility department monitors the use of the assessment list and its necessary evolution to meet the technological or regulatory changes. It updates the standards or internal policies on AI systems and ensures that the use of such systems complies with the current legal and regulatory framework and to the values of the organisation.
Product and Service Development or equivalent	The Product and Service Development department uses the assessment list to evaluate AI-based products and services and logs all the results. These results are discussed at management level, which ultimately approves the new or revised AI-based applications.
Quality Assurance	The Quality Assurance department (or equivalent) ensures and checks the results of the assessment list and takes action to escalate an issue higher up if the result is not satisfactory or if unforeseen results are detected.
HR	The HR department ensures the right mix of competences and diversity of profiles for developers of AI systems. It ensures that the appropriate level of training is delivered on Trustworthy AI inside the organisation.
Procurement	The procurement department ensures that the process to procure AI-based products or services includes a check of Trustworthy AI.
Day-to-day Operations	Developers and project managers include the assessment list in their daily work and document the results and outcomes of the assessment.

▪ *Using the Trustworthy AI assessment list*

When using the assessment list in practice, we recommend paying attention not only to the areas of concern but also to the questions that cannot be (easily) answered. One potential problem might be the lack of diversity of skills and competences in the team developing and testing the AI system, and therefore it might be necessary to involve other stakeholders inside or outside the organisation. It is strongly recommended to log all results both in technical terms and in management terms, ensuring that the problem solving can be understood at all levels in the governance structure.

This assessment list is meant to guide AI practitioners to achieve Trustworthy AI. The assessment should be tailored to the specific use case in a proportionate way. During the piloting phase, specific sensitive areas might be revealed and the need for further specifications in such cases will be evaluated in the next steps. While this

⁵⁵ <https://www.mckinsey.com/business-functions/operations/our-insights/secrets-of-successful-change-implementation>

⁵⁶ See for instance A. Bryson, E. Barth and H. Dale-Olsen, The Effects of Organisational change on worker well-being and the moderating role of trade unions, *ILRRReview*, 66(4), July 2013; Jirjahn, U. and Smith, S.C. (2006). 'What Factors Lead Management to Support or Oppose Employee Participation—With and Without Works Councils? Hypotheses and Evidence from Germany's Industrial Relations, 45(4), 650–680; Michie, J. and Sheehan, M. (2003). 'Labour market deregulation, "flexibility" and innovation', *Cambridge Journal of Economics*, 27(1), 123–143.

assessment list does not provide concrete answers to address the raised questions, it encourages reflection on how Trustworthy AI can be operationalised, and on the potential steps that should be taken in this regard.

- *Relation to existing law and processes*

It is also important for AI practitioners to recognise that there are various existing laws mandating particular processes or prohibiting particular outcomes, which may overlap and coincide with some of the measures listed in the assessment list. For example, data protection law sets out a series of legal requirements that must be met by those engaged in the collection and processing of personal data. Yet, because Trustworthy AI also requires the ethical handling of data, internal procedures and policies aimed at securing compliance with data protection laws might also help to facilitate ethical data handling and can hence complement existing legal processes. Compliance with this assessment list is *not*, however, evidence of legal compliance, nor is it intended as guidance to ensure compliance with applicable laws.

Moreover, many AI practitioners already have existing assessment tools and software development processes in place to ensure compliance also with non-legal standards. The below assessment should not necessarily be carried out as a stand-alone exercise, but can be incorporated into such existing practices.

TRUSTWORTHY AI ASSESSMENT LIST (PILOT VERSION)

1. Human agency and oversight

Fundamental rights:

- ✓ Did you carry out a fundamental rights impact assessment where there could be a negative impact on fundamental rights? Did you identify and document potential trade-offs made between the different principles and rights?
- ✓ Does the AI system interact with decisions by human (end) users (e.g. recommended actions or decisions to take, presenting of options)?
 - Could the AI system affect human autonomy by interfering with the (end) user's decision-making process in an unintended way?
 - Did you consider whether the AI system should communicate to (end) users that a decision, content, advice or outcome is the result of an algorithmic decision?
 - In case of a chat bot or other conversational system, are the human end users made aware that they are interacting with a non-human agent?

Human agency:

- ✓ Is the AI system implemented in work and labour process? If so, did you consider the task allocation between the AI system and humans for meaningful interactions and appropriate human oversight and control?
 - Does the AI system enhance or augment human capabilities?
 - Did you take safeguards to prevent overconfidence in or overreliance on the AI system for work processes?

Human oversight:

- ✓ Did you consider the appropriate level of human control for the particular AI system and use case?
 - Can you describe the level of human control or involvement?
 - Who is the "human in control" and what are the moments or tools for human intervention?
 - Did you put in place mechanisms and measures to ensure human control or oversight?
 - Did you take any measures to enable audit and to remedy issues related to governing AI autonomy?
- ✓ Is there is a self-learning or autonomous AI system or use case? If so, did you put in place more specific mechanisms of control and oversight?
 - Which detection and response mechanisms did you establish to assess whether something could go wrong?

- Did you ensure a stop button or procedure to safely abort an operation where needed? Does this procedure abort the process entirely, in part, or delegate control to a human?

2. Technical robustness and safety

Resilience to attack and security:

- ✓ Did you assess potential forms of attacks to which the AI system could be vulnerable?
 - Did you consider different types and natures of vulnerabilities, such as data pollution, physical infrastructure, cyber-attacks?
- ✓ Did you put measures or systems in place to ensure the integrity and resilience of the AI system against potential attacks?
- ✓ Did you verify how your system behaves in unexpected situations and environments?
- ✓ Did you consider to what degree your system could be dual-use? If so, did you take suitable preventative measures against this case (including for instance not publishing the research or deploying the system)?

Fallback plan and general safety:

- ✓ Did you ensure that your system has a sufficient fallback plan if it encounters adversarial attacks or other unexpected situations (for example technical switching procedures or asking for a human operator before proceeding)?
- ✓ Did you consider the level of risk raised by the AI system in this specific use case?
 - Did you put any process in place to measure and assess risks and safety?
 - Did you provide the necessary information in case of a risk for human physical integrity?
 - Did you consider an insurance policy to deal with potential damage from the AI system?
 - Did you identify potential safety risks of (other) foreseeable uses of the technology, including accidental or malicious misuse? Is there a plan to mitigate or manage these risks?
- ✓ Did you assess whether there is a probable chance that the AI system may cause damage or harm to users or third parties? Did you assess the likelihood, potential damage, impacted audience and severity?
 - Did you consider the liability and consumer protection rules, and take them into account?
 - Did you consider the potential impact or safety risk to the environment or to animals?
 - Did your risk analysis include whether security or network problems such as cybersecurity hazards could pose safety risks or damage due to unintentional behaviour of the AI system?
- ✓ Did you estimate the likely impact of a failure of your AI system when it provides wrong results, becomes unavailable, or provides societally unacceptable results (for example discrimination)?
 - Did you define thresholds and did you put governance procedures in place to trigger alternative/fallback plans?
 - Did you define and test fallback plans?

Accuracy

- ✓ Did you assess what level and definition of accuracy would be required in the context of the AI system and use case?
 - Did you assess how accuracy is measured and assured?
 - Did you put in place measures to ensure that the data used is comprehensive and up to date?
 - Did you put in place measures in place to assess whether there is a need for additional data, for example to improve accuracy or to eliminate bias?
- ✓ Did you verify what harm would be caused if the AI system makes inaccurate predictions?
- ✓ Did you put in place ways to measure whether your system is making an unacceptable amount of inaccurate predictions?
- ✓ Did you put in place a series of steps to increase the system's accuracy?

Reliability and reproducibility:

- ✓ Did you put in place a strategy to monitor and test if the AI system is meeting the goals, purposes and intended applications?
 - Did you test whether specific contexts or particular conditions need to be taken into account to ensure reproducibility?
 - Did you put in place verification methods to measure and ensure different aspects of the system's reliability and reproducibility?
 - Did you put in place processes to describe when an AI system fails in certain types of settings?
 - Did you clearly document and operationalise these processes for the testing and verification of the reliability of AI systems?
 - Did you establish mechanisms of communication to assure (end-)users of the system's reliability?

3. Privacy and data governance

Respect for privacy and data Protection:

- ✓ Depending on the use case, did you establish a mechanism allowing others to flag issues related to privacy or data protection in the AI system's processes of data collection (for training and operation) and data processing?
- ✓ Did you assess the type and scope of data in your data sets (for example whether they contain personal data)?
- ✓ Did you consider ways to develop the AI system or train the model without or with minimal use of potentially sensitive or personal data?
- ✓ Did you build in mechanisms for notice and control over personal data depending on the use case (such as valid consent and possibility to revoke, when applicable)?
- ✓ Did you take measures to enhance privacy, such as via encryption, anonymisation and aggregation?
- ✓ Where a Data Privacy Officer (DPO) exists, did you involve this person at an early stage in the process?

Quality and integrity of data:

- ✓ Did you align your system with relevant standards (for example ISO, IEEE) or widely adopted protocols for daily data management and governance?
- ✓ Did you establish oversight mechanisms for data collection, storage, processing and use?
- ✓ Did you assess the extent to which you are in control of the quality of the external data sources used?
- ✓ Did you put in place processes to ensure the quality and integrity of your data? Did you consider other processes? How are you verifying that your data sets have not been compromised or hacked?

Access to data:

- ✓ What protocols, processes and procedures did you follow to manage and ensure proper data governance?
 - Did you assess who can access users' data, and under what circumstances?
 - Did you ensure that these persons are qualified and required to access the data, and that they have the necessary competences to understand the details of data protection policy?
 - Did you ensure an oversight mechanism to log when, where, how, by whom and for what purpose data was accessed?

4. Transparency

Traceability:

- ✓ Did you establish measures that can ensure traceability? This could entail documenting the following methods:
 - Methods used for designing and developing the algorithmic system:
 - Rule-based AI systems: the method of programming or how the model was built;
 - Learning-based AI systems; the method of training the algorithm, including which input data was gathered and selected, and how this occurred.

- Methods used to test and validate the algorithmic system:
 - Rule-based AI systems; the scenarios or cases used in order to test and validate;
 - Learning-based model: information about the data used to test and validate.
- Outcomes of the algorithmic system:
 - The outcomes of or decisions taken by the algorithm, as well as potential other decisions that would result from different cases (for example, for other subgroups of users).

Explainability:

- ✓ Did you assess:
 - to what extent the decisions and hence the outcome made by the AI system can be understood?
 - to what degree the system's decision influences the organisation's decision-making processes?
 - why this particular system was deployed in this specific area?
 - what the system's business model is (for example, how does it create value for the organisation)?
- ✓ Did you ensure an explanation as to why the system took a certain choice resulting in a certain outcome that all users can understand?
- ✓ Did you design the AI system with interpretability in mind from the start?
 - Did you research and try to use the simplest and most interpretable model possible for the application in question?
 - Did you assess whether you can analyse your training and testing data? Can you change and update this over time?
 - Did you assess whether you can examine interpretability after the model's training and development, or whether you have access to the internal workflow of the model?

Communication:

- ✓ Did you communicate to (end-)users – through a disclaimer or any other means – that they are interacting with an AI system and not with another human? Did you label your AI system as such?
- ✓ Did you establish mechanisms to inform (end-)users on the reasons and criteria behind the AI system's outcomes?
 - Did you communicate this clearly and intelligibly to the intended audience?
 - Did you establish processes that consider users' feedback and use this to adapt the system?
 - Did you communicate around potential or perceived risks, such as bias?
 - Depending on the use case, did you consider communication and transparency towards other audiences, third parties or the general public?
- ✓ Did you clarify the purpose of the AI system and who or what may benefit from the product/service?
 - Did you specify usage scenarios for the product and clearly communicate these to ensure that it is understandable and appropriate for the intended audience?
 - Depending on the use case, did you think about human psychology and potential limitations, such as risk of confusion, confirmation bias or cognitive fatigue?
- ✓ Did you clearly communicate characteristics, limitations and potential shortcomings of the AI system?
 - In case of the system's development: to whoever is deploying it into a product or service?
 - In case of the system's deployment: to the (end-)user or consumer?

5. Diversity, non-discrimination and fairness

Unfair bias avoidance:

- ✓ Did you establish a strategy or a set of procedures to avoid creating or reinforcing unfair bias in the AI system, both regarding the use of input data as well as for the algorithm design?
 - Did you assess and acknowledge the possible limitations stemming from the composition of the used data sets?
 - Did you consider diversity and representativeness of users in the data? Did you test for specific populations or problematic use cases?
 - Did you research and use available technical tools to improve your understanding of the data, model and performance?